x^2 Factorizing quadratic equations

SOME QUADRATIC EQUATIONS (EQUATIONS IN THE FORM $AX^2 + BX + C = 0$) CAN BE SOLVED BY FACTORIZING.

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The quadratic formula 192–193 ❯

Quadratic factorization

Factorization is the process of finding the terms that multiply together to form another term. A quadratic equation is factorized by rearranging it into two bracketed parts, each containing a variable and a number. To find the values in the parentheses, use the rules from multiplying parentheses (see p.176)—that the numbers add together to give b and multiply together to give c of the original quadratic equation.

a is a number that multiplies x^2

b is a number that multiplies x

$$ax^2 + bx + c = 0$$

$x^2 = x \times x$

can also be a minus sign

c is a number by itself

parentheses set next to each other are multiplied together

$$(x + ??)(x + ??) = 0$$

these two unknown numbers add together to give b and multiply together to give c of the original equation

△ **A quadratic equation**
All quadratic equations have a squared term (x^2), a term that is multiplied by x, and a numerical term. The letters a, b, and, c all stand for different numbers.

△ **Two parentheses**
A quadratic equation can be factorized as two parentheses, each with an x and a number. Multiplied out, they result in the equation.

Solving simple quadratic equations

To solve quadratic equations by factorization, first find the missing numerical terms in the parentheses. Then solve each one separately to find the answers to the original equation.

To solve a quadratic equation, first look at its b and c terms. The terms in the two parentheses will need to add together to give b (6 in this case) and multiply together to give c (8 in this case).

x^2 means $1x^2$

c term is 8

$$x^2 + 6x + 8 = 0 \quad \Rightarrow \quad (x + ??)(x + ??) = 0$$

b term is 6

answer is always 0

these two numbers add together to give 6 and multiply together to give 8

To find the unknown terms, draw a table. In the first column, list the possible combinations of numbers that multiply together to give the value of c = 8. In the second column, add these terms together to see if they add up to b = 6.

list possible factors of c=8

add factors to find their sum

Two numbers that multiply to give c=8	The sum of the two numbers	
8 and 1	$8 + 1 = 9$	✗
4 and 2	$4 + 2 = 6$	✓

all sets of numbers in this column multiply to give c=8

not all the factors add up to produce the needed sum (b), which is 6

4 and 2 are the factors needed, as 4 + 2 = 6, which is b

Insert the factors into the parentheses after the x terms. Because the two parentheses multiplied together equal the original quadratic expression, they can also be set to equal 0.

$$(x + ??)(x + ??) = 0 \quad \Rightarrow \quad (x + 4)(x + 2) = 0$$

insert first value here

insert second value here

For the two parentheses to multiply to equal 0, the value of either one needs to be 0. Set each one equal to 0 and solve. The resulting values are the two solutions of the original equation.

solve for first value

subtract 4 from both sides to isolate x

$$x + 4 = 0 \quad \Rightarrow \quad x = -4$$

one possible solution is –4

solve for second value

subtract 2 from both sides to isolate x

$$x + 2 = 0 \quad \Rightarrow \quad x = -2$$

another possible solution is –2